

Square-difference-free sets of exponent 0.75806746

Ordered intervals, composite digit codes, and a binary recursion

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September 18, 2026

*This paper was written by GPT-6-Astra under my prompting and supervision. I recommend that readers explore the paper through a dialogue with AI, rather than a traditional reading. Upload the paper to your preferred AI assistant and ask questions such as: “What are the key ideas that improve on the previous best bound?”, “How, exactly do those ideas work?” and “Where is the meat of the argument?” Follow up whenever an explanation leaves something unclear. The goal is to ask the paper questions, and to understand the mathematics and the contribution through conversation, rather than directly reading the text.

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Abstract

Let $D(N)$ be the largest cardinality of a subset of $\{1, \dots, N\}$ containing no nonzero square difference. We give a computer-assisted construction proving $D(N) \geq N^{0.75806746 - o(1)}$. This improves the exponent 0.752796455874514... in Krachun’s ranked-block construction. Our local objects are residue sets carrying intervals ordered along modular square differences. A finite stopping-word argument shows that a surplus of interval-moment contributions produces an asymptotic lower bound. We combine six prime-chain alphabets, two composite alphabets with three restricted digits in each prime coordinate, and a 25-state binary recursion. All finite witnesses and the exact verification programs are embedded in the $\text{\LaTeX}$ source and attached to this PDF. The proof includes the lifting and recursion arguments, the stopping-word criterion, and the justification of the finite checking procedures. The numerical conclusion is computer-assisted; the accompanying Lean 4 development formalizes the full bound, including all finite certificate conditions.

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1 Introduction

1.1 The problem and the quantitative landscape

For a positive integer N , put

$$D(N) = \max\{|A| : A \subseteq \{1, \dots, N\}, \quad (A - A) \cap \{1, 4, 9, \dots\} = \emptyset\}.$$

Furstenberg and Sárközy independently proved that $D(N) = o(N)$ [1, 2]. The quantitative upper-bound problem has subsequently developed through the work of Pintz, Steiger and Szemerédi, Bloom and Maynard, and Green and Sawhney [4, 11, 12]. In particular, Green and Sawhney prove a bound of the form $D(N) \ll N \exp(-c\sqrt{\log N})$ for an absolute $c > 0$ [12, Theorem 1.1]. Our concern is the complementary construction problem: to maximize α in $D(N) \geq N^{\alpha-o(1)}$. All logarithms in this paper are natural. The displayed notation means that, for every $\varepsilon > 0$, the bound $D(N) \geq N^{\alpha-\varepsilon}$ holds for every sufficiently large N .

Ruzsa's construction gives exponent

$$\frac{1}{2} + \frac{\log 7}{2 \log 65} = 0.733077 \dots$$

by restricting alternating base-65 digits and leaving the others free [3]. Beigel and Gasarch, and independently Lewko, use a 12-element square-difference-free residue set modulo 205 to obtain

$$\frac{1}{2} + \frac{\log 12}{2 \log 205} = 0.7334117970 \dots \quad [5, 6].$$

For this construction family, a conjectural restriction on the size of square-difference-free sets modulo square-free moduli gives a natural three-quarter barrier; it is not an upper bound on $D(N)$ [3, 6]. Further bounds for square-avoiding sets modulo square-free integers are studied by Gabdullin [9]. These address local square avoidance, whereas the local objects here permit directed square edges. Computational exploration by Georgiev, Gómez-Serrano, Tao and Wagner recovered the modulus-205 construction without improving its exponent [13, Section 16]; this records a search outcome, not an optimality theorem.

In the function-field analogue, Naslund [7, Conjecture 13] conjectured that square-difference-free sets of polynomials of degree below n over $\mathbb{F}_q$ have size at most $q^{3n/4}$, for odd q and $4 \mid n$. Jones [8] disproves this at $q = 3$ for every $n \geq 8$ divisible by 4, with a Lean-verified construction registered in Palomar.

Krachun crossed the integer three-quarter barrier by allowing modular square differences, provided that they consistently decrease a rank, and then combining the ranks over coprime moduli [14]. His exponent is

$$\alpha_K = \frac{10 + \sum_{i=1}^{10} \log p_i / \log t_i}{1 + 2 \sum_{i=1}^{10} \log p_i / \log t_i} = 0.752796455874514 \dots, \quad (1)$$

where

$$(p_i, t_i) = (3, 2), (7, 3), (11, 4), (19, 5), (23, 5), (31, 7), (43, 7), (59, 9), (71, 9), (103, 11).$$

Here t_i is the size of a specified transitive sequence in the Paley tournament modulo p_i . Bounds for such transitive subtournaments explain some limitations of this particular prime-chain family [10, 14]; those limitations do not apply to all ordered interval alphabets.

Theorem 1.1. *Let $\alpha_0 = 37903373/50000000 = 0.75806746$. For every $\varepsilon > 0$ there is $N_0(\varepsilon)$ such that, for every integer $N \geq N_0(\varepsilon)$, there exists $A \subseteq \{1, \dots, N\}$ with no nonzero square difference and*

$$|A| \geq N^{\alpha_0 - \varepsilon}.$$

Relative to (1), the increase in the exponent is approximately 0.005271004125486. The comparison concerns the published/preprint construction just described, not an upper bound on the true optimum. Intermediate computations in the working research archive reached 0.7580419 and 0.758043893; these are stages of the present research, not additional bibliographic antecedents. The certificates used here were newly saved and do not depend on any missing historical computational files.

1.2 How the improvement works

The ranked-block realization and CRT gluing steps are already present in Krachun’s Lemmas 4 and 5 [14]. We prove them below with the opposite rank convention: a modular square difference increases the rank, while the chosen integer representative decreases with rank. The additional flexibility has three concrete sources.

First, a digit is assigned an interval $[a_x, a_x + w_x]$ rather than just a rank. A square edge $x \rightarrow y$ requires $a_x + w_x \leq a_y$, but incomparable digits may share substantial portions of their intervals. Unequal widths permit nonuniform digit frequencies. The useful statistic is the moment $\sum_x w_x^f$, rather than just the number of digits or the unweighted height of their graph. This is closely related to weighted chain constraints in a partially ordered set [15] and to the familiar entropy–log-sum identity [16]; neither theory is needed as a black box in our proof.

Second, the odd components are built modulo 215^6 and 437^6 , grouping primes as $215 = 5 \cdot 43$ and $437 = 19 \cdot 23$. Each prime coordinate has three restricted low digits, interleaved with free digits. Which position first differs is determined separately at each prime. Conditional choices of higher digits and changes to child supports reduce unnecessary interval constraints. We certify the resulting explicit supports directly, so their search histories are unnecessary for the proof.

Third, the prime 2 supplies a component through a recursion whose states record separate windows for even and odd digits. Four residue branches, reflection, and parity changes allow a useful moment growth rate. The recursion uses only 25 states and 94 branches, although the final alphabet is taken at the fixed depth 10^{10} . It is represented by induction, never by an explicit list of all its digits.

The common principle is an interval-moment criterion. If alphabets in pairwise coprime square bases b_i satisfy

$$\sum_{x \in C_i} w_{i,x}^{f_i} \geq b_i^\alpha, \quad f_i \geq 0, \quad \sum_i f_i > \alpha, \quad (2)$$

then α is attainable. We prove this by stopping interval words at a common width and selecting a single word length in each component. This finite argument avoids asymptotic composition estimates. For our nine components every f_i is an exact rational, and

$$\sum_i f_i - \alpha_0 = \frac{25671}{10^{12}} > 0. \quad (3)$$

1.3 Scope and self-contained computational evidence

Theorem 1.1 is a computer-assisted theorem. Its analytic and combinatorial implications are proved in the text; specific finite rational inequalities are established by the supplied executable certificates.

No optimization output, floating-point eigenvalue, or search optimality claim is an assumption of the proof.

The L^AT_EX source contains a complete Python attachment, including all witnesses and checking programs. Compiling that one source file writes the attachment and embeds it in the PDF. Section 7 specifies its data and acceptance conditions, and Appendix A gives extraction and execution instructions. Thus neither an external research note nor an online data download is needed to reconstruct the construction. The large raw integer arrays are distributed as an attachment, not as thousands of typeset rows. Standard Python 3 and ordinary exact integer/rational arithmetic suffice for verification.

2 Ranked residues and ordered interval alphabets

For $b \geq 1$, let $\mathcal{Q}(b) = \{z^2 \bmod b : z \in \mathbb{Z}\}$, including zero. Residues are represented canonically in $\{0, \dots, b-1\}$. For distinct x, y write $x \rightarrow_b y$ if $y - x \in \mathcal{Q}(b)$. For composite b , a nonzero square residue may vanish in some prime coordinates; invertibility is never imposed.

Definition 2.1. A ranked alphabet modulo b is a finite set C of canonical residues with an integer function $r : C \rightarrow \{0, \dots, h-1\}$, for some $h \geq 1$, such that $x \rightarrow_b y$ implies $r(x) < r(y)$. An ordered interval alphabet assigns to every $x \in C$ an interval $[a_x, a_x + w_x] \subseteq [0, 1]$, with $w_x > 0$, such that $x \rightarrow_b y$ implies $a_x + w_x \leq a_y$. Its moment of order $f \geq 0$ is $Z_f(C, w) = \sum_{x \in C} w_x^f$.

Intervals may overlap when the corresponding digits are incomparable. The graph of an ordered interval alphabet is acyclic, since a directed cycle would imply that the sum of its positive widths is at most zero.

Lemma 2.2 (Path budgets). *Let a finite directed acyclic graph have positive vertex weights w_x . There are intervals in $[0, 1]$ of those widths ordered along every edge if and only if the sum of the weights on every directed path is at most one.*

Proof. Ordered intervals along a path have disjoint interiors, so their widths sum to at most one. Conversely, let d_x be the maximum weight of a path ending at x , including x . Set $a_x = d_x - w_x$. Then $a_x \geq 0$ and $a_x + w_x = d_x \leq 1$. If $x \rightarrow y$, appending y to a maximizing path ending at x gives $d_y \geq d_x + w_y$, whence $a_y \geq a_x + w_x$. $\square$

This is the weighted-chain viewpoint underlying the chain polytope of a finite poset [15]; the elementary proof above supplies the part used here.

Lemma 2.3 (Reverse-rank realization). *A ranked alphabet of size q modulo M , with ranks in $\{0, \dots, h-1\}$, gives a square-difference-free set of size q in $\{1, \dots, Mh\}$.*

Proof. For the canonical residue x , choose

$$T(x) = 1 + x + M(h - 1 - r(x)).$$

These numbers lie in the stated interval and are distinct modulo M . If $T(y) - T(x) = z^2 > 0$, then $x \neq y$ and reduction modulo M implies $r(y) - r(x) \geq 1$. Consequently

$$T(y) - T(x) = y - x - M(r(y) - r(x)) \leq (M - 1) - M < 0,$$

a contradiction. $\square$

Lemma 2.4 (CRT gluing). *Suppose $M_1, \dots, M_n$ are pairwise coprime and C_i is a ranked alphabet modulo M_i , with q_i digits and ranks below h_i . Their CRT product is a ranked alphabet modulo $M = \prod_i M_i$, with $Q = \prod_i q_i$ digits and ranks below*

$$h = 1 + \sum_{i=1}^n (h_i - 1).$$

Proof. The Chinese remainder map identifies residues modulo M with tuples of residues modulo the M_i . Explicitly, writing $P_i = M/M_i$, choose u_i with $P_i u_i \equiv 1 \pmod{M_i}$ and take the canonical residue of $\sum_i x_i P_i u_i$. This proves existence; divisibility by every M_i proves uniqueness. Assign rank $r(x) = \sum_i r_i(x_i)$. A square difference modulo M is a square in each coordinate. Equal coordinates leave the rank unchanged and unequal ones increase it. At least one coordinate is unequal, so the sum strictly increases. The size and rank bounds follow. $\square$

Lemmas 2.3 and 2.4 express the ranked realization and gluing mechanism of [14, Lemmas 4–5] in our convention.

Lemma 2.5 (Square-base words). *Let $b = s^2 > 1$. If digits C form a ranked alphabet with rank height h , then their length- k words form a ranked alphabet modulo b^k with $|C|^k$ digits and rank height h^k . If the digits instead carry ordered intervals, the word intervals obtained by affine composition are ordered along modular square differences as well.*

Proof. Encode a word by $X = \sum_{j=0}^{k-1} x_j b^j$. Uniqueness of the base- b expansion gives injectivity. Suppose $Y - X \equiv z^2 \pmod{b^k}$ and $X \neq Y$. Let j be the first position at which $x_j \neq y_j$, reading from the least significant digit. Since $s^{2j} = b^j$ divides z^2 , we have $s^j \mid z$: this follows by comparing the valuations of every prime dividing s . Dividing the congruence by b^j and reducing modulo b gives $y_j - x_j \equiv (z/s^j)^2 \pmod{b}$, a nonzero square edge.

In the ranked case assign the word rank $R(X) = \sum_{j=0}^{k-1} r(x_j) h^{k-1-j}$. The first differing digit increases its rank, and its place value dominates every possible decrease in the remaining places. Thus $R(Y) > R(X)$. Its range is $0 \leq R < h^k$.

For intervals, put $F_x(t) = a_x + w_x t$ and assign $F_{x_0} \circ \dots \circ F_{x_{k-1}}([0, 1])$. At the first differing position, the two digit intervals are ordered. The remaining compositions lie inside those digit intervals, and the common preceding maps are increasing. This preserves the required order. The width is $W_X = \prod_{j=0}^{k-1} w_{x_j}$. $\square$

Lemma 2.6 (Rounding intervals to ranks). *If every interval in an ordered alphabet has width at least $\gamma > 0$, then $r(x) = \lfloor a_x / \gamma \rfloor$ is a valid rank with $0 \leq r(x) < \lceil 1/\gamma \rceil$.*

Proof. On an edge $a_y \geq a_x + \gamma$, so $r(y) \geq r(x) + 1$. Moreover $0 \leq a_x \leq 1 - \gamma$, which gives the stated upper bound. $\square$

3 A finite interval-moment criterion

Theorem 3.1. *Let $n \geq 1$. For each $i \in \{1, \dots, n\}$ let C_i be an ordered interval alphabet in a square base $b_i > 1$, with the bases pairwise coprime. Suppose there are common constants $0 < \sigma \leq \rho < 1$ such that $\sigma \leq w_{i,x} \leq \rho$ for all digits. If $\alpha \geq 0$, $f_i \geq 0$, and*

$$Z_{f_i}(C_i, w_i) \geq b_i^\alpha \quad (1 \leq i \leq n), \quad F := \sum_{i=1}^n f_i > \alpha,$$

then $D(N) \geq N^{\alpha - o(1)}$.

The common width bounds are automatic for a fixed finite collection of nonempty alphabets with every width strictly between zero and one. They need not be uniform over a sequence of proposed constructions.

Lemma 3.2 (Stopping and selecting a length). *Fix one alphabet in base b , widths $\sigma \leq w_x \leq \rho < 1$, and $f \geq 0$. Put $q = b^\alpha$ and suppose $Z = \sum_x w_x^f \geq q > 0$. For every integer $K \geq 1$, with $\delta = \rho^K$, there exist $1 \leq k \leq K$ and a set D of length- k words such that*

$$\sigma\delta < W_X \leq \delta \quad (X \in D), \quad q^k \leq (K+1)|D|\delta^f.$$

Proof. Start with the empty word of width one. Extend each active word by all digits; a child is active if its width exceeds δ , and terminal otherwise. Only active words are extended. No active word survives depth K , since all depth- K widths are at most $\rho^K = \delta$. A terminal word has a parent wider than δ , so its own width is strictly greater than $\sigma\delta$.

Let A_j be the sum of W^f over active depth- j words, and let T_j be the sum over terminal words newly created at depth $j+1$. Multiplication of widths gives the exact identities

$$A_0 = 1, \quad A_K = 0, \quad A_{j+1} + T_j = ZA_j \quad (0 \leq j < K).$$

If $T_j < q^{j+1}/(K+1)$ for every $j < K$, then induction, using $Z \geq q$, gives

$$A_j \geq \frac{K+1-j}{K+1} q^j \quad (0 \leq j \leq K).$$

Indeed, subtracting T_j from ZA_j proves the next inequality, strictly when $j+1 > 0$. At $j = K$ this contradicts $A_K = 0$. Choose a j for which $T_j \geq q^{j+1}/(K+1)$ and retain precisely those terminal words; set $k = j+1$. Since $f \geq 0$ and $W \leq \delta$, their moment is at most $|D|\delta^f$. This proves the claim. $\square$

Proof of Theorem 3.1. Apply Lemma 3.2 separately in each component with the same K and $\delta = \rho^K$. Obtain lengths $k_i \in [1, K]$ and word sets D_i . Their moduli $b_i^{k_i}$ are coprime perfect squares, and Lemmas 2.5 and 2.6 give ranks below $H = \lceil 1/(\sigma\delta) \rceil$ in every component. Put $M = \prod_i b_i^{k_i}$ and $Q = \prod_i |D_i|$. Lemma 2.4 gives a ranked alphabet of size Q modulo M with height $h = 1 + n(H-1)$. Multiplying the selected-moment estimates yields

$$M^\alpha \leq Q\delta^F(K+1)^n. \tag{4}$$

Since $\delta \leq 1$, the fixed constant $c = (n+1)(1+\sigma^{-1})$ satisfies $h \leq c/\delta$. Consequently

$$(Mh)^\alpha \leq Qc^\alpha(K+1)^n\delta^{F-\alpha}.$$

Because $F-\alpha > 0$ and $0 < \rho < 1$, the last factor $c^\alpha(K+1)^n\rho^{K(F-\alpha)}$ tends to zero. For example, the ratio of consecutive terms tends to $\rho^{F-\alpha} < 1$, which proves this without any asymptotic counting formula. Choose and fix K so that $(Mh)^\alpha \leq Q$.

Write $L = Mh > 1$. Repeating this one ranked alphabet t times and applying Lemmas 2.5 and 2.3 gives Q^t square-difference-free integers in $[1, L^t]$. For any $N \geq 1$, let t be the integer with $L^t \leq N < L^{t+1}$. Then

$$N^\alpha \leq L^{(t+1)\alpha} \leq Q^{t+1}, \quad D(N) \geq Q^t \geq Q^{-1}N^\alpha.$$

For every $\varepsilon > 0$, $N^\varepsilon \geq Q$ eventually, giving $D(N) \geq N^{\alpha-\varepsilon}$ for every sufficiently large N . All alphabets, K , and hence L, Q have been fixed before N varies. $\square$

Remark 3.3 (Relation to entropy). The criterion also explains the search objective. For a positive probability vector π on the digits, define $H(\pi) = -\sum_x \pi_x \log \pi_x$ and $R(\pi) = -\sum_x \pi_x \log w_x$. With $\nu_x = w_x^f / Z_f$, the inequality $\log u \leq u - 1$ gives $\sum_x \pi_x \log(\nu_x / \pi_x) \leq 0$. Thus

$$H(\pi) - fR(\pi) \leq \log Z_f,$$

with equality when $\pi = \nu$. This is the entropy variational identity [16]. Equal-width chains are a special case, whereas unequal widths exploit the actual partial order. The stopping argument proves the criterion directly and needs neither this identity nor Stirling's formula.

4 Odd-prime digit constructions

4.1 Prime chains

Lemma 4.1. *Let $p \equiv 3 \pmod{4}$ be prime and let $(s_0, \dots, s_{t-1})$ be distinct residues such that $s_j - s_i$ is a nonzero square modulo p whenever $i < j$. Then the pt digits $s_j + pu$, $0 \leq u < p$, form an ordered interval alphabet modulo p^2 , with interval $[j/t, (j+1)/t]$ on each digit $s_j + pu$. Its moment is pt^{1-f} .*

Proof. The element -1 is not a square modulo p : otherwise an element of order four would exist in the multiplicative group of order $p-1$, contradicting $4 \nmid p-1$. A square difference between different low digits therefore points from the earlier to the later chain entry. If the low digits agree, a nonzero difference has valuation one, and cannot be a square modulo p^2 . The intervals are ordered in the first case and no edge is possible in the second. There are p free copies of each interval, giving the moment. $\square$

This is the local Paley-chain mechanism of [14]; the alternating free-digit principle goes back to [3].

4.2 Separate first differences at different primes

Fix distinct odd primes $p_1, \dots, p_s$ and positive integers $e_1, \dots, e_s$. A low point a consists of words $(a_{p,0}, \dots, a_{p,e_p-1})$ with $0 \leq a_{p,j} < p$. For distinct low points a, b , impose an edge $a \rightsquigarrow b$ if, for every prime p , either the coordinate words are identical or their first differing position j satisfies

$$b_{p,j} - a_{p,j} \in \mathcal{Q}(p) \setminus \{0\}. \quad (5)$$

The first differing position is allowed to depend on p . No comparison is required from a prime whose entire low word is unchanged.

Lemma 4.2 (Prime-coordinate free-digit lift). *Suppose a finite set S of distinct low points has positive intervals in $[0, 1]$ ordered along every edge $\rightsquigarrow$. Put $F = \prod_p p^{e_p}$. For each $a \in S$, choose arbitrary $u_{p,j} \in \{0, \dots, p-1\}$ and form*

$$x_p = \sum_{j=0}^{e_p-1} (a_{p,j} p^{2j} + u_{p,j} p^{2j+1}) \pmod{p^{2e_p}}. \quad (6)$$

Combining the x_p by CRT produces an ordered interval alphabet in base F^2 , with $F|S|$ digits. Every free copy uses its low point's interval, and the moment is $F \sum_{a \in S} w_a^f$.

Proof. Unique prime-adic digit expansions and CRT show that all these digits are distinct and count their number. Suppose different lifted digits x, y have a square difference modulo F^2 . Consider a prime coordinate in which they differ. If $k < 2e_p$ is the first differing actual digit position, then $y_p - x_p$ is divisible by p^k but not by p^{k+1} : all lower digits agree and the coefficient at p^k is nonzero modulo p . The congruence to a square forces k to be even. Indeed, if $z^2 \equiv y_p - x_p \pmod{p^{2e_p}}$, the valuation of z^2 below $2e_p$ must equal k , so $k = 2j$; dividing shows that the leading nonzero difference is a square modulo p .

If the low coordinate words differ, the first actual difference must thus occur at their first differing low digit, and (5) holds. If two distinct lifted digits have the same low point in every coordinate, some first actual difference would instead lie in a free, odd position, a contradiction. Thus their low points are distinct, and each coordinate is either identical as a low word or obeys (5). The assumed low interval order therefore gives the required lifted order. Counting free copies gives the moment formula. $\square$

A single common first-difference index across all prime coordinates would impose extra edges that this proof does not require. This distinction lets us use conditional higher low digits and different child supports. Only the final set of points, not a proposed recursive description of how it was discovered, is an input to Lemma 4.2.

5 A binary recursion with parity windows

5.1 States and transitions

A state s has an even window E_s and an odd window O_s in $[0, 1]$. A window may be absent; each present window is a closed interval of positive length, and at least one window is present. At depth m , the state will describe an ordered interval alphabet modulo 4^m , with the interval of each digit inside the window of its parity.

A transition from s has at most one branch for each $r \in \{0, 1, 2, 3\}$. A branch specifies a child j , scale $u > 0$, shift $t \in \mathbb{R}$, and bits $\tau, \eta \in \{0, 1\}$, called swap and reflect. For a child digit x modulo $Q = 4^{m-1}$, let

$$x' = ((-1)^\eta x + \tau) \bmod Q, \quad y = r + 4x'. \quad (7)$$

When $\eta = 1$ reflect the child interval $[a, a + w]$ to $[1 - a - w, 1 - a]$; then apply the increasing affine map $z \mapsto t + uz$. The parity of x' is the parity of x toggled by τ , since Q is even. For transformed parity $p \in \{0, 1\}$, let J_{r+4p} be the placed window obtained from the child window of parity p xor τ , with the same reflection and affine map. An absent child window contributes no J_{r+4p} .

The transition conditions are

$$J_{r+4p} \subseteq E_s \quad (r \text{ even}), \quad J_{r+4p} \subseteq O_s \quad (r \text{ odd}), \quad (8)$$

$$\sup J_k \leq \inf J_{k+1 \bmod 8} \quad \text{whenever both windows are present.} \quad (9)$$

In particular, (9) includes the comparison from class 7 to class 0. Missing classes make such a cyclic specification feasible.

Lemma 5.1 (Binary transition). *For $m \geq 2$, if the child states are valid at depth $m - 1$ and (8)–(9) hold, the branches construct a valid state at depth m .*

Proof. Different branches give different residues modulo four, and within a branch (7) is injective. Thus the digits are distinct. Containment of intervals follows from (8).

Suppose $Y - X$ is a nonzero square modulo 4^m . If the digits belong to one branch, then $Y - X = 4(y' - x')$ modulo 4^m . Any corresponding square root is even, so division by four gives a square difference modulo Q . Without reflection the child order applies directly. With reflection the transformed difference equals the negative of the original child difference; consequently the child square edge has the opposite orientation. Reflection of the intervals reverses that order once more, giving the required transformed interval order. Translation cancels in differences, and the positive scale preserves order.

For different branches whose residues differ by two modulo four, the difference is 2 modulo four and is not a square. In the remaining case the difference is odd. Every odd square is 1 modulo eight, since $(2k + 1)^2 = 4k(k + 1) + 1$. Therefore the numerical classes of X, Y are $k, k + 1$ cyclically modulo eight. Their intervals lie in the corresponding J windows, so (9) orders them. This exhausts all cases. Only the necessary condition on odd squares is used. $\square$

5.2 Moment growth and a finite depth

Fix $0 < f < 1$. Let $v_s > 0$ and $a > 1$ satisfy

$$\sum_{\text{branches } s \rightarrow j} u^f v_j \geq a v_s \quad \text{for every } s. \quad (10)$$

This positive-vector certificate is a familiar device for bounding the growth of a nonnegative matrix [17]. Here the required consequence is just the following elementary induction.

Lemma 5.2. *Let d_s be the length of the wider present window of state s , and put $c = \min_s d_s / v_s > 0$. The transitions construct depth- m states with moments $Z_{s,m} \geq c a^{m-1} v_s$ for all $m \geq 1$.*

Proof. At depth one choose just the digit 0 or 1 of the parity of the wider window and assign it the whole window. A singleton has no nontrivial square edge. Its moment d_s^f is at least d_s , since $0 < d_s \leq 1$. This gives $Z_{s,1} \geq c v_s$. Reflection and translation do not change widths, scaling multiplies each moment by u^f , and different branches are disjoint. Thus

$$Z_{s,m} = \sum_{s \rightarrow j} u^f Z_{j,m-1} \geq c a^{m-2} \sum_{s \rightarrow j} u^f v_j \geq c a^{m-1} v_s.$$

Validity follows inductively from Lemma 5.1. $\square$

If a root state has both windows $[0, 1]$ and $v_{\text{root}} = 1$, it produces an unrestricted interval alphabet modulo 4^m . Shrinking every final interval at its left endpoint by a factor of $1/2$ preserves all orders, makes all widths at most $1/2$, and multiplies the moment by 2^{-f} . Therefore the condition

$$\log c + (m - 1) \log a - f \log 2 - \alpha m \log 4 > 0 \quad (11)$$

implies the required moment $Z_f > (4^m)^\alpha$. This uses only a fixed finite depth, even if its numerical value is enormous.

6 The explicit nine components

We now fix all the data used in Theorem 1.1. Every terminating decimal in the construction and in its verification thresholds denotes an exact rational. The six chains are as follows; all displayed forward differences are nonzero squares modulo the indicated prime.

p	t	Ordered chain
3	2	0, 1
7	3	0, 1, 2
11	4	0, 1, 4, 5
31	7	0, 1, 8, 5, 2, 9, 10
59	9	0, 1, 28, 17, 22, 4, 26, 20, 29
103	11	0, 1, 29, 92, 2, 61, 17, 30, 93, 18, 19

There are 244 ordered pairs of distinct entries in these six rows. The finite checker verifies the square relation in both orientations for all of them. Lemma 4.1 gives the corresponding prime alphabets.

Component	Square base b_i	Exact moment power f_i
Chain at 3	3^2	0.181945506487
Chain at 7	7^2	0.085799249226
Chain at 11	11^2	0.107233269134
Chain at 31	31^2	0.089166212567
Chain at 59	59^2	0.042173711274
Chain at 103	103^2	0.002397851691
Three-low-digit code at 5, 43	215^6	0.026537312662
Three-low-digit code at 19, 23	437^6	0.067871875356
Binary root, $m = 10^{10}$	$4^{10^{10}}$	0.154942497274

Table 1: The parameters used throughout the proof.

For a chain of length t modulo p , the real number $f_p^* = 1 - (2\alpha_0 - 1) \log p / \log t$ would give equality $pt^{1-f_p^*} = p^{2\alpha_0}$. The table uses a rational rounded down to twelve decimal places instead. Exact logarithm bounds certify that each displayed f_p lies in $(0, f_p^*)$ and hence gives the strict moment inequality $pt^{1-f_p} > p^{2\alpha_0}$. The following are conservative rational lower bounds for $(1 - f_p) \log t - (2\alpha_0 - 1) \log p$:

$p = 3$	$p = 7$	$p = 11$	$p = 31$	$p = 59$	$p = 103$
$1.98 \cdot 10^{-13}$	$7.60 \cdot 10^{-14}$	$8.09 \cdot 10^{-13}$	$3.99 \cdot 10^{-13}$	$2.21 \cdot 10^{-13}$	$2.06 \cdot 10^{-12}$

6.1 The two odd witnesses

The attachment contains two explicit finite arrays. Their mathematical format is described in Section 7; these arrays are part of the construction, not parameters to be supplied later.

	215 component	437 component
Primes	5, 43	19, 23
Low digits per prime	3	3
Low points	4,913	19,683
Directed low edges	6,031,254	77,614,486
Width denominator	10^{14}	10^{16}
Free multiplicity	215^3	437^3

For the 215 component, integer longest-path computation in the supplied order gives maximum path weight

$$\frac{99999974386028}{10^{14}} < 1.$$

Lemma 2.2 therefore constructs the interval starts from the listed widths. For the 437 component the starts are supplied explicitly; every directed edge satisfies its interval inequality and the largest right endpoint is

$$\frac{9999999524038902}{10^{16}} < 1.$$

All widths are positive and strictly smaller than one. The low-point moment inequalities have the certified margins

$$\log \sum_{a \in S_{215}} w_a^{f_{215}} - (2\alpha_0 - 1) \log(215^3) > 6.97840 \cdot 10^{-10}, \quad (12)$$

$$\log \sum_{a \in S_{437}} w_a^{f_{437}} - (2\alpha_0 - 1) \log(437^3) > 6.95598 \cdot 10^{-10}. \quad (13)$$

Lemma 4.2 multiplies the moments by the free multiplicities, so the lifted alphabets satisfy $Z_{f_i} > b_i^{\alpha_0}$.

6.2 The binary witness

The attachment specifies 25 states, 94 branches and a positive rational vector v . State indices start at zero; the root has index 1, both its windows are $[0, 1]$, and $v_1 = 1$. The exact parameters are

$$f_B = 0.154942497274, \quad a = \frac{1430118728343}{500000000000}, \quad m = 10^{10}, \quad c = \frac{285516532458}{803818168703}.$$

The finite geometry test checks 304 containment and cyclic-order inequalities, including the wrap from class 7 to class 0. It also checks all state, branch, and positivity requirements. The growth-row and logarithmic margins are

$$\min_s \left(\sum_{s \rightarrow j} u^{f_B} v_j - a v_s \right) > 6.99293 \cdot 10^{-9}, \quad (14)$$

$$\log a - \alpha_0 \log 4 > 3.05203 \cdot 10^{-9}, \quad (15)$$

$$\log c + (m - 1) \log a - f_B \log 2 - \alpha_0 m \log 4 > 28.32697028. \quad (16)$$

The rational c is checked by taking the minimum of d_s/v_s over the 25 states. Lemmas 5.1 and 5.2, followed by final halving of widths, give $Z_{f_B} > (4^m)^{\alpha_0}$.

6.3 Closing the bound

Proof of Theorem 1.1. The exact finite checks just specified are established by the executable certificate and the checking procedures proved sound in Section 7. Lemmas 4.1, 4.2, 5.1 and 5.2 turn the finite data into nine ordered interval alphabets satisfying the moments in Table 1. All these alphabets are finite and nonempty, and all widths are in $(0, 1)$. In particular, the huge binary depth is fixed now. Taking the minimum and maximum over the finitely many widths gives common $0 < \sigma \leq w \leq \rho < 1$; no useful numerical lower bound on σ is needed.

The prime supports are disjoint: the six chain primes, the pairs $\{5, 43\}$ and $\{19, 23\}$, and the prime 2. Thus the nine bases are pairwise coprime perfect squares. Adding the exact rational powers gives

$$\sum_p f_p = 0.508715800379, \quad \sum_i f_i = 0.758067485671 = \alpha_0 + \frac{25671}{10^{12}}.$$

Every hypothesis of Theorem 3.1 now holds with $\alpha = \alpha_0$. That theorem supplies the conclusion for every sufficiently large N , with all finite choices made before the limit. $\square$

7 Exact finite verification

7.1 Data sufficient to define the construction

The embedded archive contains the manifest and the three witness JSON files listed in Appendix B, together with their checking programs. The manifest fixes α_0 , both odd powers, the prime chains, and the SHA-256 hashes of the three witnesses. The binary witness fixes its own power, depth, vector and growth parameter. The rational chain check gives the six powers in Table 1. All parameters used by the proof are compared exactly; numerical fields left over from discovery are ignored.

For an odd witness, **primes** is the ordered list of coordinate primes and **depth** is the number of low digits in each coordinate. The array **points** lists the low words; each entry consists of one length-three word per prime, in increasing significance. A common positive integer **width_denominator** D and positive integers **width_numerators** n_j define $w_j = n_j/D$. The 437 file additionally gives **start_numerators** s_j , defining $a_j = s_j/D$. The 215 starts are defined by Lemma 2.2. Thus the listed data specify every interval without reference to the optimization programs.

In the binary witness, a state's window record is $[a, b, c, d, q]$ and means $E_s = [a/q, b/q]$, $O_s = [c/q, d/q]$; the pair $c = d = -1$ denotes an absent odd window. Every present window is checked to lie in $[0, 1]$ with positive length. Each branch lists its residue **r**, **child** index, **swap** and **reflect** bits, and rational **scale** and **shift** as numerator–denominator pairs. The vector entries are rational strings. Equations (7)–(10) specify precisely what this data means.

7.2 Complete geometric checks

The 215 checker checks prime validity, distinctness of points, all digit ranges and word lengths, positivity of the rational widths, and all ordered distinct pairs. For each pair it evaluates (5) separately in each coordinate. Every edge must point forward in the supplied list. Writing d_j for a path numerator, it then computes

$$d_j = n_j + \max(\{d_i : i \rightsquigarrow j\} \cup \{0\})$$

and checks $\max_j d_j \leq D$. This is exactly the recurrence in Lemma 2.2; induction in the checked topological order proves its validity. No claim about a transitive reduction is used.

For the larger 437 witness the checker computes the identical full edge relation with integer bitsets. Here is why this accelerated check is complete. For one prime, let $G(v)$ be the bitset of source points whose coordinate word begins with the prefix v . For a target word $b = (b_0, \dots, b_{e-1})$, its allowable sources have bitset

$$G(b) \cup \bigcup_{j=0}^{e-1} \bigcup_{q \in \mathcal{Q}(p) \setminus \{0\}} G((b_0, \dots, b_{j-1}, b_j - q) \bmod p). \quad (17)$$

The first term covers identical words; each remaining term fixes the first difference and permits all subsequent digits. Thus (17) is exactly the relation required at that prime. Intersect the bitsets over primes and remove the target itself. This enumerates every incoming edge. The checker also verifies that every incoming index precedes its target.

The rational inequalities are checked without iterating through individual bits. Sort vertices by end numerator $s_i + n_i$, sweep target starts in increasing order, and maintain the bitset B of vertices whose ends exceed the current start. The incoming bitset must have empty intersection with B . At each step B has exactly that definition, so this is equivalent to checking $s_i + n_i \leq s_j$ on every edge $i \rightsquigarrow j$. All sizes, ranges, distinctness, starts, widths and endpoints are checked separately. The counts in Section 6 are outputs of these complete checks, not sampled tests.

The binary geometry checker constructs every J window with exact fractions and checks (8) and (9), alongside unique branch residues, valid child indices, positive scales, and valid bits. Its 304 comparisons count two endpoint containments per present placed window and one inequality per present adjacent pair. Lemma 5.1 proves why these finitely many conditions imply validity at arbitrary depth. A direct small-depth test is not a substitute for that lemma and is unnecessary for the main verification.

7.3 Rational logarithm enclosures

All decisive numerical comparisons use rational bounds. We give the details both to specify their meaning and to separate verification from discovery. For $1 \leq x \leq 2$, put $z = (x-1)/(x+1) \in [0, 1/3]$. Integration of the geometric series, or subtraction of the series for $\log(1+z)$ and $\log(1-z)$, gives

$$\log x = 2 \sum_{j=0}^{n-1} \frac{z^{2j+1}}{2j+1} + R_n, \quad 0 \leq R_n \leq \frac{2z^{2n+1}}{(2n+1)(1-z^2)}. \quad (18)$$

The tail bound follows by replacing all its denominators by $2n+1$ and summing the resulting geometric series. For any positive rational x , write $x = 2^k y$ with $1 \leq y \leq 2$ and use $\log x = k \log 2 + \log y$. If $k < 0$, the interval endpoints for $k \log 2$ are reversed. Every series term and bound is rational.

The 215 checker uses 32 terms of (18) with exact fractions, and the binary checker uses 70. The 437 checker uses 48 terms with outward rounding at scale $S = 2^{192}$. More explicitly, if $L_z \leq Sz \leq U_z$, set

$$L_{z^2} = \lfloor L_z^2/S \rfloor, \quad U_{z^2} = \lceil U_z^2/S \rceil.$$

Starting from L_z, U_z , multiply these bounds successively, rounding every lower product down and every upper product up. In summing the j th term, round division by $2j+1$ in the same directions. After n terms the upper bound U_{next} for Sz^{2n+1} gives the additional scaled tail bound

$$\left\lceil \frac{2U_{\text{next}}S}{(2n+1)(S - U_{z^2})} \right\rceil.$$

Here $S - U_{z^2} > 0$, as follows from $z \leq 1/3$ and the displayed scale. Induction preserves enclosure through every operation, proving that the resulting interval contains the exact logarithm. These are standard outward-rounding principles [18], with the specific error bound proved above.

For clarity, write the resulting rational endpoints as $\ell(x), u(x)$, so $\ell(x) \leq \log x \leq u(x)$. A positive rational proposal L for x^f is accepted as a lower bound only when

$$u(L) \leq f\ell(x), \quad f \geq 0. \quad (19)$$

Then $\log L \leq f \log x$, and hence $L \leq x^f$. Similarly a proposed upper endpoint U is accepted only if $\ell(U) \geq fu(x)$. Decimal arithmetic proposes candidates; it never decides the validity of a power enclosure. A poor proposal causes rejection rather than a false positive.

For the odd components the proposals are integer multiples of 10^{-24} . If $L_j \leq w_j^f$ have been accepted, the checker proves

$$\ell\left(\sum_j L_j\right) - (2\alpha_0 - 1)e \sum_p u(p) > 0.$$

Since $2\alpha_0 - 1 > 0$, this proves the lifted moment inequality. Lower bounds for the margins in (12)–(13) are obtained from that same rational expression. For the binary component, accepted lower powers give lower bounds on $\sum u^f v_j - av_s$. The fixed-depth check uses

$$\ell(c) + (m - 1)\ell(a) - f_B u(2) - \alpha_0 m u(4) > 0.$$

This proves (16) directly, without approximating the astronomically large alphabet. The chain check proves $(1 - f_p)\ell(t) - (2\alpha_0 - 1)u(p) > 0$ for each of the six rational powers. Finally (3) is an exact rational addition.

7.4 Independent arithmetic and the trust boundary

An additional program in the attachment checks the nontrivial moment, binary growth, and global arithmetic using a different expansion:

$$\log x = \sum_{k=1}^n \frac{t^k}{k} + R'_n, \quad t = 1 - 1/x \in [0, 1/2], \quad 0 \leq R'_n \leq \frac{t^{n+1}}{(n+1)(1-t)}$$

for $1 \leq x \leq 2$. It uses 230 terms and outward-rounded integer bounds with scale 10^{60} ; its power proposals use denominator 10^{40} . The same geometric-tail and induction arguments prove its enclosures. This check does not replace the geometry checks. It provides independently implemented arithmetic for the same pinned witnesses.

The complete supplied run accepts all the finite conditions above. This is executable finite evidence: the correctness of the numerical theorem uses execution of the supplied integer/rational programs, not just their printed success messages. The elementary implications of the conditions have been proved in the text. The code can be inspected independently of its compressed storage by extracting the attachment.

A separate Lean 4 development proves Theorem 3.1 and the full numerical conclusion $D(N) \geq N^{0.75806746-o(1)}$, including all finite certificate conditions. It formalizes the six prime-chain components, both reconstructed composite alphabets, and the entire binary recursion, together with their order and numerical moment checks. The proof uses only the usual foundational axioms; no finite computation remains an unproved hypothesis. Lean and Mathlib provide the proof infrastructure [19, 20]. The English argument above is complete on its own and uses no Lean theorem as a premise.

8 What the construction leaves open

The proof isolates finite quantities that future searches can improve: feasible low-point supports and their interval moments, the choice of coprime prime groups, and the binary transition geometry. Increasing local cardinality can be unhelpful if it forces more ordering or a worse width profile. Likewise, a child support that performs worse on its own can be useful when its interfaces to other

children reduce the combined ordering cost. Such observations motivate searches; the theorem depends only on the final certified arrays.

The number 0.75806746 is a certified feasible exponent, not an asserted optimum for the method or for $D(N)$. In particular, the possible limitations of prime Paley chains discussed in [14, 10] do not bound the composite and binary interval alphabets used here. The present proof gives no construction of exponent 0.76 and no new general upper bound. Its very large fixed depths also make it unsuitable as a practical construction for moderate N ; the objective here is the asymptotic exponent.

Acknowledgements and provenance. OpenAI Codex agents assisted with mathematical exploration, certificate construction and checking, Lean formalization, literature checking, and manuscript preparation. The elementary ranked-block and free-digit ideas are credited above to their mathematical antecedents. The present proof and saved certificates consolidate the September 2026 working research; no endorsement by the cited authors is asserted.

A Reproducing the entire finite verification

The attachment is a self-contained Python program carrying a compressed archive of the exact data and all programs needed by the checks.

[square-difference-free-certificate.py](#)

A PDF viewer supporting embedded files can save this attachment. Alternatively, compiling the single `square-difference-free-sets-of-exponent-0.75806746.tex` source with `pdflatex` writes that same Python file beside the build outputs. The source contains the actual archive bytes, not a link to them. Run

```
python3 square-difference-free-certificate.py
```

The program checks the embedded archive hash, extracts into a temporary directory, runs the complete geometry/moment verifier, runs the rational chain-power check, and runs the independent arithmetic check. It makes no network requests and needs no third-party Python packages. To retain the extracted data, source code and newly generated reports, choose a fresh output directory:

```
python3 square-difference-free-certificate.py --extract certificate-check
```

The run finishes with

```
PAPER CERTIFICATE PASS: complete geometry, rational target powers, and independent
arithmetic.
```

The wrapper and complete entry points reject Python's `-O` and `-OO` modes, because some constituent routines use assertions as checks. Use the supplied entry point rather than bypassing those guards. Compilation of the PDF does not execute these mathematical checks; they are a separate explicit verification step.

The principal verification files in the extracted archive are:

- `research-notes/improved-bound/verify_bound.py`: complete manifest, component, chain, coprimality and global-budget checking;
- `research-notes/odd-search/verify_candidate.py`: complete pairwise low-code and longest-path checking, exact rational logarithms;

- `research-notes/odd-push/verify_fast.py`: equivalent full-edge bitset checking and directed fixed-point logarithms;
- `research-notes/binary-search/check_geometry.py` and `research-notes/binary-search/certify.py`: binary transition geometry, positive-vector inequalities and fixed-depth moment;
- `research-notes/lean-planning/rational_chain_powers.py`: the six rational powers and the exact surplus in Table 1;
- `research-notes/binary-push/independent_arithmetic.py`: independent arithmetic using the second logarithm series.

The archive preserves the historical file names for reproducibility; they do not introduce dependencies on any other files in the working repository. Discovery-only functions present in some source modules are not executed.

B Certificate identity

The following SHA-256 hashes fix the exact bytes of the data files in the embedded archive. Hashes identify witnesses; the finite checks, rather than the hashes themselves, establish their mathematical properties.

Manifest: `research-notes/capacity-bound/bound.json`

836b231c32e771dfaf494a74de40b5213c8f8606865ac8309724e6a1fc5ec491

Odd 215: `research-notes/odd-search/q215-depth3-product.json`

dae38a756de83c99d65973ff85f23a7068ce6e85a7a4f12400a923f24cf46574

Odd 437: `research-notes/odd-push/q437-depth3-final.json`

7abd0c007c42fa1a5484ea8a4d02c56b04898209f74a0403c5503c03395e58d3

Binary: `research-notes/binary-push/certificate-alpha-0758067460.json`

4835df59b0cc8444abe274a2c0d923fa89d023687c52ec58a7c31a49a8b03c87

The SHA-256 of the embedded ZIP archive is

5dcaffc0c826e178ad9b7ec3b42007f92eb4f7f44a03fe9c24012247f87b1e6b.

The attached program checks this identity before unpacking.

References

- [1] H. Furstenberg, *Ergodic behavior of diagonal measures and a theorem of Szemerédi on arithmetic progressions*, J. Analyse Math. **31** (1977), 204–256. doi:10.1007/BF02813304.
- [2] A. Sárközy, *On difference sets of sequences of integers. I*, Acta Math. Acad. Sci. Hungar. **31** (1978), no. 1–2, 125–149. doi:10.1007/BF01896079.
- [3] I. Z. Ruzsa, *Difference sets without squares*, Period. Math. Hungar. **15** (1984), no. 3, 205–209. doi:10.1007/BF02454169.

- [4] J. Pintz, W. L. Steiger, and E. Szemerédi, *On sets of natural numbers whose difference set contains no squares*, J. London Math. Soc. (2) **37** (1988), no. 2, 219–231. doi:[10.1112/jlms/s2-37.2.219](https://doi.org/10.1112/jlms/s2-37.2.219).
- [5] R. Beigel and W. Gasarch, *Square-difference-free sets of size $\Omega(n^{0.7334\dots})$* , 2008, [arXiv:0804.4892v3](https://arxiv.org/abs/0804.4892v3).
- [6] M. Lewko, *An improved lower bound related to the Furstenberg–Sárközy theorem*, Electron. J. Combin. **22** (2015), no. 1, Paper 1.32, 6 pp. doi:[10.37236/4656](https://doi.org/10.37236/4656).
- [7] E. Naslund, *Paley graphs and Sárközy’s theorem in function fields*, 2022, [arXiv:2203.01293v3](https://arxiv.org/abs/2203.01293v3), Conjecture 13.
- [8] JD Jones, *ns-lean: Naslund’s Conjecture 13 is false at $q = 3$, $k = 2$, for every n congruent to 0 mod 4 with $n \geq 8$* , Palomar, entry [PALOMAR-2026-09-17-000003](https://palomar.com/entry/PALOMAR-2026-09-17-000003), version 1, 17 September 2026.
- [9] M. R. Gabdullin, *Sets in $\mathbb{Z}_m$ whose difference sets avoid squares*, Sb. Math. **209** (2018), no. 11, 1603–1610. doi:[10.1070/SM8992](https://doi.org/10.1070/SM8992).
- [10] S. Satake, *On the restricted isometry property of the Paley matrix*, Linear Algebra Appl. **631** (2021), 35–47. doi:[10.1016/j.laa.2021.08.018](https://doi.org/10.1016/j.laa.2021.08.018).
- [11] T. F. Bloom and J. Maynard, *A new upper bound for sets with no square differences*, Compos. Math. **158** (2022), no. 8, 1777–1798. doi:[10.1112/S0010437X22007679](https://doi.org/10.1112/S0010437X22007679).
- [12] B. Green and M. Sawhney, *New bounds for the Furstenberg–Sárközy theorem*, 2025, [arXiv:2411.17448v2](https://arxiv.org/abs/2411.17448v2) (first version 2024).
- [13] B. Georgiev, J. Gómez-Serrano, T. Tao, and A. Z. Wagner, *Mathematical exploration and discovery at scale*, 2025, [arXiv:2511.02864v1](https://arxiv.org/abs/2511.02864v1).
- [14] D. Krachun, *Square-difference-free sets beyond the three-quarter barrier*, 2026, [arXiv:2608.01325v1](https://arxiv.org/abs/2608.01325v1).
- [15] R. P. Stanley, *Two poset polytopes*, Discrete Comput. Geom. **1** (1986), 9–23. doi:[10.1007/BF02187680](https://doi.org/10.1007/BF02187680).
- [16] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, second ed., Wiley-Interscience, Hoboken, NJ, 2006. doi:[10.1002/047174882X](https://doi.org/10.1002/047174882X).
- [17] E. Seneta, *Non-negative Matrices and Markov Chains*, second ed., Springer Series in Statistics, Springer, New York, 1981, reprinted 2006. doi:[10.1007/0-387-32792-4](https://doi.org/10.1007/0-387-32792-4).
- [18] R. E. Moore, R. B. Kearfott, and M. J. Cloud, *Introduction to Interval Analysis*, Society for Industrial and Applied Mathematics, Philadelphia, PA, 2009. doi:[10.1137/1.9780898717716](https://doi.org/10.1137/1.9780898717716).
- [19] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in *Automated Deduction—CADE 28*, A. Platzer and G. Sutcliffe (eds.), Lecture Notes in Computer Science **12699**, Springer, Cham, 2021, 625–635. doi:[10.1007/978-3-030-79876-5_37](https://doi.org/10.1007/978-3-030-79876-5_37).
- [20] The mathlib Community, *The Lean mathematical library*, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, Association for Computing Machinery, New York, 2020, 367–381. doi:[10.1145/3372885.3373824](https://doi.org/10.1145/3372885.3373824).